

C# Beginner's Essential Reference

1. Basic Data Types

Category	Data Type	Description	Example
Integer Numbers	int	Whole numbers (32-bit)	int age = 25;
	long	Large whole numbers (64-bit)	long distance = 100000L;
Decimal Numbers	float	Single-precision (7 digits)	float price = 9.99f;
	double	Double-precision (15-16 digits)	double pi = 3.14159;
	decimal	Precision for financial (28-29 digits)	decimal salary = 5000.50m;
Text	char	Single character	char grade = 'A';
	string	Text (immutable)	string name = "Alice";
Boolean	bool	True/false	bool isActive = true;

2. Operators

Arithmetic Operators

Operator	Description	Example
+	Addition	int sum = 5 + 3;
-	Subtraction	int diff = 5 - 3;
*	Multiplication	int prod = 5 * 3;
/	Division	double div = 5 / 3.0;
%	Modulus (remainder)	int rem = 5 % 3;
++	Increment	age++;
--	Decrement	count--;

Comparison Operators

Operator	Description	Example
==	Equal to	if (age == 18)
!=	Not equal	if (name != "Bob")
>	Greater than	if (score > 90)
<	Less than	if (temp < 0)
>=	Greater or equal	if (items >= 10)
<=	Less or equal	if (price <= 100)

Logical Operators

Operator	Description	Example
&&	AND	if (age > 18 && hasLicense)
	OR	if (isStudent isSenior)
!	NOT	if (!isComplete)

Assignment Operators

Operator	Shortcut For	Example
=	Assign	int x = 5;
+=	x = x + y	x += 3;
-=	x = x - y	x -= 2;
*=	x = x * y	x *= 4;
/=	x = x / y	x /= 2;

3. Essential Methods

Integer (int) Operations & Methods

Method/Operator	Description	Example
ToString()	Converts an integer to a string	string numStr = num.ToString();
Parse()	Converts a string to an integer	int num = int.Parse("123");
TryParse()	Safely converts a string to an integer	bool success = int.TryParse("123", out int num);
CompareTo()	Compares two integers	int result = num1.CompareTo(num2);
Math.Abs()	Gets the absolute value	int absValue = Math.Abs(-5);
Math.Max()	Returns the larger of two integers	int max = Math.Max(3, 5);
Math.Min()	Returns the smaller of two integers	int min = Math.Min(3, 5);
Math.Pow()	Raises a number to a power (returns double)	double power = Math.Pow(2, 3);
++ (Increment)	Increases value by 1	num++; or ++num;

Method/Operator	Description	Example
-- (Decrement)	Decreases value by 1	num--; or --num;
+ (Addition)	Adds two integers	int sum = num1 + num2;
- (Subtraction)	Subtracts two integers	int diff = num1 - num2;
* (Multiplication)	Multiplies two integers	int product = num1 * num2;
/ (Division)	Divides two integers (truncates)	int quotient = num1 / num2;
% (Modulus)	Gets the remainder of division	int remainder = num1 % num2;
Bitwise Operators	Performs bitwise operations (&, , ^, ~, <<, >>)	int result = num1 & num2;

String Methods & Operations

Method/Property	Description	Example
Length	Gets the number of characters	int len = str.Length;
ToUpper()	Converts to uppercase	string upper = str.ToUpper();
ToLower()	Converts to lowercase	string lower = str.ToLower();
Trim()	Removes leading/trailing whitespace	string trimmed = str.Trim();
TrimStart()	Trims leading whitespace	string trimmed = str.TrimStart();
TrimEnd()	Trims trailing whitespace	string trimmed = str.TrimEnd();
Substring()	Extracts part of a string	string sub = str.Substring(2, 5);
Replace()	Replaces a substring	string newStr = str.Replace("old", "new");
Contains()	Checks if substring exists	bool exists = str.Contains("word");
StartsWith()	Checks if string starts with substring	bool starts = str.StartsWith("Hi");

Method/Property	Description	Example
EndsWith()	Checks if string ends with substring	<code>bool ends = str.EndsWith(".");</code>
IndexOf()	Gets the first index of a substring	<code>int pos = str.IndexOf("abc");</code>
LastIndexOf()	Gets the last index of a substring	<code>int pos = str.LastIndexOf("xyz");</code>
Split()	Splits into an array by delimiter	<code>string[] parts = str.Split(',');</code>
Join() (static)	Combines an array into a string	<code>string joined = string.Join("-", arr);</code>
Concat() (static)	Concatenates strings	<code>string combined = string.Concat(s1, s2);</code>
Format() (static)	Formats a string (like "\$")	<code>string formatted = string.Format("{0} {1}", "Hello", "World");</code>
IsNullOrEmpty()	Checks if null or empty	<code>bool empty = string.IsNullOrEmpty(str);</code>
IsNullOrWhiteSpace()	Checks if null/whitespace	<code>bool blank = string.IsNullOrWhiteSpace(str);</code>
PadLeft()	Pads left with spaces (or custom char)	<code>string padded = str.PadLeft(10, '-');</code>
PadRight()	Pads right with spaces (or custom char)	<code>string padded = str.PadRight(10);</code>
ToCharArray()	Converts to a char[]	<code>char[] chars = str.ToCharArray();</code>
Equals()	Checks equality (case-sensitive)	<code>bool same = str1.Equals(str2);</code>
Compare() (static)	Compares strings lexicographically	<code>int result = string.Compare(s1, s2);</code>
Interpolation (\$)	Embeds variables in a string	<code>string msg = \$"Hello {name}!";</code>
Escape Sequences		

Math Methods

Method	Description	Example
Math.Max(x, y)	Returns larger value	int max = Math.Max(5, 10); → 10
Math.Min(x, y)	Returns smaller value	int min = Math.Min(5, 10); → 5
Math.Round(x)	Rounds to nearest integer	double r = Math.Round(3.6); → 4
Math.Abs(x)	Absolute value	int pos = Math.Abs(-5); → 5

Array/List Methods

Method	Description	Example
Length	Get array size	int size = arr.Length;
Sort()	Sort an array	Array.Sort(numbers);
Add()	Add to a List<T>	names.Add("Alice");
Remove()	Remove from a List<T>	names.Remove("Bob");

Dictionary Methods

Method	Description	Example
Add()	Adds a key-value pair to the dictionary	dict.Add("key", "value");
Remove()	Removes an entry by key	dict.Remove("key");
ContainsKey()	Checks if a key exists	bool exists = dict.ContainsKey("key");
ContainsValue()	Checks if a value exists	bool exists = dict.ContainsValue("value");
Clear()	Removes all entries	dict.Clear();
Keys	Gets all keys in the dictionary	var keys = dict.Keys;
Values	Gets all values in the dictionary	var values = dict.Values;
TryGetValue()	Safely gets a value by key	dict.TryGetValue("key", out string val);
Count	Gets the number of key-value pairs	int size = dict.Count;

4. Type Conversion

Method	Description	Example
Convert.ToInt32(x)	Convert to int	int num = Convert.ToInt32("123");
Convert.ToDouble(x)	Convert to double	double d = Convert.ToDouble("3.14");
x.ToString()	Convert to string	string s = age.ToString();
int.Parse(x)	Parse string to int (throws error if invalid)	int n = int.Parse("42");